

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         int[] arr = {10, 20, 30, 40, 50};
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the index: ");
9         int index = sc.nextInt();
10
11         if (index >= 0 && index < arr.length) {
12             System.out.println("Element at index " + index + " is " + arr[index]
13                 );
14         } else {
15             System.out.println("Invalid index!");
16         }
17
18         sc.close();
19     }
20 }
```

Enter the index: 1
Element at index 1 is 20

=== Code Execution Successful ===

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 20, 30, 40, 50};
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter element to search: ");
        int key = sc.nextInt();

        int low = 0, high = arr.length - 1;
        while (low <= high) {
            int mid = (low + high) / 2;
            if (arr[mid] == key) {
                System.out.println("Element found at index " + mid);
                return;
            } else if (key < arr[mid]) {
                high = mid - 1;
            } else {
                low = mid + 1;
            }
        }
        System.out.println("Element not found");
    }
}

```

Enter element to search: 20
Element found at index 1

=== Code Execution Successful ===

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 20, 30, 40, 50};
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the index: ");
        int index = sc.nextInt();

        if (index >= 0 && index < arr.length) {
            System.out.println("Element at index " + index + " is " + arr[index]);
        } else {
            System.out.println("Invalid index!");
        }

        sc.close();
    }
}
```

Enter the index: 1
Element at index 1 is 20

=== Code Execution Successful ===

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int max = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] > max) {
                max = arr[i];
            }
        }

        System.out.println("Maximum element = " + max);
    }
}

```

Enter the number of elements: 5
Enter the array elements:

8

9

4

5

6

Maximum element = 9

=== Code Execution Successful ===3

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {1, 2, 3, 4};  
  
        for (int i = 0; i < arr.length; i++) {  
            for (int j = i + 1; j < arr.length; j++) {  
                System.out.println("(" + arr[i] + ", " + arr[j] + ")");  
            }  
        }  
    }  
}
```

(1, 2)
(1, 3)
(1, 4)
(2, 3)
(2, 4)
(3, 4)

=== Code Execution Successful ===